

4-2 Milestone Three: Enhancement Two: Algorithms and Data Structure

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This artifact is a C++ bid-sorting application originally developed in 2023 for CS-300: Data Structures and Algorithms. The program loads bid data from a CSV file, stores the records in a vector, and allows users to sort the data using different sorting algorithms. I selected this artifact for my ePortfolio because it provides a strong demonstration of my understanding of algorithms, data structures, and performance analysis. The original project implemented both selection sort and quick sort, but for this enhancement I redesigned the quick sort implementation into a hybrid sorting algorithm that uses insertion sort for smaller partitions. This improvement reduces recursive overhead and increases efficiency while preserving the original functionality of the application. I also implemented a comparison counter to provide measurable performance data beyond execution time alone.

This enhancement aligns with Course Outcome 3 by demonstrating my ability to design, evaluate, and improve computing solutions through the application of algorithmic principles and performance-based decision making. The project required analyzing the strengths and weaknesses of the original implementation, identifying opportunities for optimization, and balancing trade-offs between complexity and efficiency. Through this process, I gained a deeper understanding of how algorithms behave in real-world scenarios and how small implementation decisions can have a meaningful impact on performance. One of the biggest challenges was improving the algorithm without changing the overall structure of the program, requiring careful integration of insertion sort into the existing quick sort framework. Overall, this enhancement reinforced the importance of continuous optimization and demonstrated that developing effective

software involves not only creating functional solutions but also evaluating and improving them to achieve better performance and scalability.